

RUTH SHARON RAJU

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EDUCATION

RUTGERS, THE STATE UNIVERSITY OF NEW JERSEY

MS in Data Science (Statistics)

NEW BRUNSWICK, NEW JERSEY, USA

Dec 2025

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (CBIT)

BE. Information Technology

HYDERABAD, INDIA

July 2023

EXPERIENCE

AI Research Engineer, *Rutgers University - Edward J. Bloustein School of Planning and Public Policy (Oct 2024 – Dec 2025)*

- Optimized medical data extraction for NIH-supported projects by engineering a sentiment analysis pipeline for 10K+ Reddit posts, achieving a 20-35% reduction in errors through the implementation of few-shot LLM setups
- Redesigned the AIXosphere website to better showcase research projects, leading to a 25% increase in session duration and improved navigation for visiting academic collaborators

Founding AI/ML Engineer, *LangPal (January 2026 - Present)*

- Architected end-to-end voice pipeline (ElevenLabs STT/TTS → Google Gemini AI → Supabase) and built fallback to handle STT misrecognitions
- Built FastAPI backend with error handling and logging; engineered Flutter mobile UI with real-time transcription and response playback; deployed to iOS and managed team code review
- Directed 5 complementary learning “games” (Match Words, Speed Vocab, Grammar Quest, etc.) with live AI-generated content, implementing quality checks to catch hallucinations in generated questions

Data Science Intern, *Transit Startup (April 2026 - Present)*

- Engineered inaccuracy index to quantify ETA prediction errors across 18 routes, time periods, and weather conditions and identified failure scenarios (6+ minute errors) and root causes through systematic analysis
- Developed break threshold detection system with statistical baselines per stop, reducing false positives in delay predictions by testing against historical data
- Built cost-optimization model reducing Google Maps API calls by 65% (\$3500/month savings) while maintaining prediction accuracy thresholds; designed A/B tests to validate that optimization didn't degrade model performance

Natural Language Autoencoders, *Personal Project (May 2026)*

- Replicated Anthropic's Natural Language Autoencoders on Qwen 1.5 7B to understand decode activations
- Extracted layer 16 activations from 20+ prompts and trained supervised reconstruction model mapping text explanations to activation patterns
- Manually authored semantic descriptions, measured reconstruction fidelity (MSE), and identified confabulation patterns in explanations; achieved 0.65 FVE norm after training

PUBLICATIONS

“Societal Impacts and Public Perception of Chatbots: Implications for Individuals and Organizations” – *Submitted to Northeast Business & Economics Association, 2025.*

Built NLP pipeline analyzing 8K+ Reddit posts (F1 0.81); deployed public Streamlit tool for chatbot sentiment analysis and bias detection; co-authored accepted conference paper (NBEA 2025)

SKILLS AND TOOLS

ML & NLP: PyTorch, Transformers, MLflow, Sentiment Analysis, Node2Vec, Prompt Engineering, Hallucination Detection

Programming: C++, Python, SQL, R

Developer Tools: GitHub, Cursor, Codex, Streamlit, FastAPI